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Blow-by gas recirculation device

Abstract

The blow-by gas recirculation device includes first to third passages and first and second check valves. The first to third passages respectively connect a portion of an intake passage downstream of a throttle valve, connect a portion of the intake passage downstream of an intercooler, and connect a portion of the intake passage upstream of a compressor to an interior of the crankcase. In the first passage, a first check valve is configured to permit a flow of gas from the interior of the crankcase to the intake passage and restrict a flow of gas from the intake passage to the interior of the crankcase. In the second passage, the second check valve is configured to permit a flow of gas from the intake passage to the interior of the crankcase and restrict a flow of gas from the interior of the crankcase to the intake passage.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0083399	12/2007	Hirano	123/572	F01M 13/02
2014/0158096	12/2013	Leone	123/563	F02B 29/0468

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110778375	12/2019	CN	C08J 7/126
202013008611	12/2013	DE	F01M 13/0011
202015100375	12/2014	DE	F01M 13/022
102014013714	12/2015	DE	F01M 13/02
102016217686	12/2017	DE	N/A
102017002218	12/2017	DE	N/A
102018212103	12/2018	DE	F01M 11/10
102019200978	12/2019	DE	F01M 1/18
S57-013814	12/1981	JP	N/A
2006-046244	12/2005	JP	N/A
2008-111422	12/2007	JP	N/A
2009299645	12/2008	JP	F01M 13/022
2017052400	12/2016	JP	N/A
2021055575	12/2020	JP	N/A
WO-2021176501	12/2020	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from prior Japanese Patent Application No. 2023-145280, filed on Sep. 7, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to a blow-by gas recirculation device.

2. Description of Related Art

(3) Japanese Laid-Open Patent Publication No. 2006-46244 describes a known blow-by gas treatment device for a forced-induction engine. The blow-by gas treatment device includes an air introduction passage that connects the portion of an intake passage downstream of a compressor and upstream of an intercooler to the interior of a crankcase. Further, the blow-by gas treatment device includes a recirculation passage that connects the portion of the intake passage upstream of the compressor to the interior of the crankcase. The blow-by gas treatment device ventilates blow-by gas during forced-induction operation of the forced-induction engine by delivering the fresh air pressurized by the compressor into the crankcase through the air introduction passage.

(4) The blow-by gas recirculation device disclosed in the above publication still has room for improvement in enhancing ventilation efficiency. For example, the disclosed blow-by gas recirculation device is unable to perform ventilation during naturally aspirated operation.

SUMMARY

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(6) A blow-by gas recirculation device according to an aspect of the present disclosure is configured to recirculate blow-by gas from an interior of a crankcase of a forced-induction engine to an intake passage. The forced-induction engine includes a compressor disposed in the intake passage, an intercooler disposed in a portion of the intake passage downstream of the compressor, and a throttle valve disposed in a portion of the intake passage downstream of the intercooler. The blow-by gas recirculation device includes a first passage that connects a portion of the intake passage downstream of the throttle valve to the interior of the crankcase, a second passage that connects the portion of the intake passage downstream of the intercooler to the interior of the crankcase, a third passage that connects a portion of the intake passage upstream of the compressor to the interior of the crankcase, a first check valve configured to permit a flow of gas from the interior of the crankcase to the intake passage through the first passage and restrict a flow of gas from the intake passage to the interior of the crankcase through the first passage, and a second check valve configured to permit a flow of gas from the intake passage to the interior of the crankcase through the second passage and restrict a flow of gas from the interior of the crankcase to the intake passage through the second passage.

(7) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram showing the configuration of a blow-by gas recirculation device according to an embodiment.

(2) FIG. 2 is a diagram illustrating the state of the blow-by gas recirculation device shown in FIG. 1 during naturally aspirated operation.

(3) FIG. 3 is a diagram illustrating the state of the blow-by gas recirculation device shown in FIG. 1 during forced-induction operation.

(4) FIG. 4 is a schematic diagram showing the configuration of a blow-by gas recirculation device according to another embodiment.

(5) Throughout the drawings and the detailed description, the same reference numerals refer to the

same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

(6) This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

(7) Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

(8) In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

(9) An embodiment of a blow-by gas recirculation device will now be described in detail with reference to FIGS. 1 to 3.

(10) Configuration of Forced-Induction Engine

(11) The configuration of a forced-induction engine **10** in which the blow-by gas recirculation device of the present embodiment is employed will now be described with reference to FIG. 1. The forced-induction engine **10** shown in FIG. 1 is a hydrogen engine that uses hydrogen as fuel. In hydrogen engines, combustible hydrogen may be included in blow-by gas. Thus, hydrogen engines require higher blow-by gas ventilation performance than gasoline or diesel engines.

(12) The forced-induction engine **10** includes a cylinder block **11**. Cylinders **12** are formed inside the cylinder block **11**. FIG. 1 shows only one of the cylinders **12**. Each cylinder **12** reciprocally accommodates a piston **13**. The portion of the cylinder **12** on the upper side of the piston **13** includes a combustion chamber **17** that burns hydrogen. An oil pan **14** that stores oil is attached to the lower part of the cylinder block **11**. The portion in the cylinder block **11** on the lower side of the cylinder **12** includes a crankcase **15**. A cylinder head **16** is mounted on the upper part of the cylinder block **11**. In the cylinder head **16**, an intake port **18** and an exhaust port **19** are individually formed for each cylinder **12**. A head cover **16A** is mounted on the upper side of the cylinder head **16**. A valve operating chamber **20** that houses a valvetrain is formed inside the upper part of the cylinder head **16**, covered by the head cover **16A**.

(13) The forced-induction engine **10** includes an intake passage **21** that introduces air into the combustion chamber **17** and an exhaust passage **22** that discharges exhaust gas from the combustion chamber **17**. The intake passage **21** includes an air cleaner **23** that filters dust or the like from the air. The portion of intake passage **21** downstream of the air cleaner **23** includes a compressor **24**. The compressor **24**, together with a turbine **25** disposed in the exhaust passage **22**, forms a turbocharger. The portion of the intake passage **21** downstream of the compressor **24** includes an intercooler **26**. The intercooler **26** is a heat exchanger used to cool the air that has been heated due to compression by the compressor **24**. The portion of the intake passage **21** downstream of the intercooler **26** includes a throttle valve **27**. The throttle valve **27** regulates the flow rate of air delivered through the intake passage **21** to the combustion chamber **17**. The intake passage **21** branches for each cylinder **12** in an intake manifold **28** that is located downstream of the throttle valve **27**. The intake manifold **28** is connected to the combustion chamber **17** through the intake port **18**.

(14) The forced-induction engine **10** includes an injector **29**, a hydrogen tank **30**, and a pressure regulator **31**. The pressure regulator **31** regulates the pressure of hydrogen in the hydrogen tank **30** and supplies it to the injector **29**. The injector **29** injects hydrogen into the air that is supplied for combustion in the combustion chamber **17**. In FIG. 1, the injector **29** is disposed to inject hydrogen into the intake port **18**. Instead, the injector **29** may be disposed to inject hydrogen into the

combustion chamber **17**.

(15) Configuration of Blow-By Gas Recirculation Device

(16) The configuration of the blow-by gas recirculation device of the present embodiment will now be described with reference to FIG. **1**. The blow-by gas recirculation device includes three passages that connect the interior of the crankcase **15** to the intake passage **21**; namely, a first passage **R1**, a second passage **R2**, and a third passage **R3**.

(17) The first passage **R1** connects the portion of the intake passage **21** downstream of the throttle valve **27** to the interior of the crankcase **15**. The first passage **R1** includes a blow-by gas passage **40**, a head-side separator **41**, a first check valve **42**, a first PCV hose **43**, and a block-side separator **44**. The head-side separator **41** and the block-side separator **44** separate oil mist from the blow-by gas flowing through the first passage **R1**. The head-side separator **41** is attached to the inner side of the head cover **16A**. The blow-by gas passage **40** runs through the interior of the cylinder block **11** and the cylinder head **16**, connecting the interior of the crankcase **15** to the head-side separator **41**. The block-side separator **44** is located in the middle of the blow-by gas passage **40** within the cylinder block **11**. The first PCV hose **43** connects the head-side separator **41** to the intake manifold **28**. The first check valve **42** permits the flow of gas from the interior of the crankcase **15** to the intake passage **21** through the first passage **R1**, and restricts the flow of gas from the intake passage **21** to the interior of the crankcase **15** through the first passage **R1**. The first check valve **42** is located in a portion of the first PCV hose **43** connected to the head-side separator **41**. In the present embodiment, the head-side separator **41** and the block-side separator **44** correspond to a first separator that separates oil mist from the blow-by gas flowing through the first passage **R1**.

(18) The second passage **R2** connects the portion of the intake passage **21** downstream of the intercooler **26** to the interior of the crankcase **15**. The second passage **R2** includes an intake-side end connected to the intake passage **21**. The second passage **R2** includes a second PCV hose **45** and a second check valve **46**. The second PCV hose **45** connects the crankcase **15** to the intake manifold **28**. The second check valve **46** permits the flow of gas from the intake passage **21** to the interior of the crankcase **15** through the second passage **R2**, and restricts the flow of gas from the interior of the crankcase **15** to the intake passage **21** through the second passage **R2**. The second check valve **46** is located at a portion of the second PCV hose **45** connected to the crankcase **15**.

(19) The third passage **R3** connects the portion of the intake passage **21** upstream of the compressor **24** to the interior of the crankcase **15**. The third passage **R3** includes an oil return passage **47**, the valve operating chamber **20**, a second separator **48**, and a third PCV hose **49**. The oil return passage **47** passes through the interior of the cylinder block **11** and the cylinder head **16**, connecting the valve operating chamber **20** to the interior of the crankcase **15**. The oil return passage **47** functions as a passage for recirculating oil from the valve operating chamber **20** to the oil pan **14**, and functions as a passage for circulating gas between the valve operating chamber **20** and the interior of the crankcase **15**. The second separator **48** separates oil mist from the blow-by gas flowing through the third passage **R3**. The second separator **48** is located on the inner side of the head cover **16A**. The third PCV hose **49** connects a portion of the intake passage **21** located downstream of the air cleaner **23** and upstream of the compressor **24** to the second separator **48**.

(20) The second passage **R2** in the blow-by gas recirculation device of the present embodiment is configured to have a minimum flow passage area that is smaller than the minimum flow passage area of the third passage **R3**. The minimum flow passage area represents a flow passage area where the flow passage area for gas is the smallest in a passage. The relationship between the minimum flow passage areas can be achieved by, for example, using a hose with a smaller diameter than the third PCV hose **49** as the second PCV hose **45**. The relationship can also be achieved by arranging a constriction in the middle of the second passage **R2**.

(21) Operation and Advantages of Present Embodiment

(22) The operation and advantages of the present embodiment will now be described with reference to FIGS. **2** and **3**. The outline arrows shown in FIGS. **2** and **3** indicate the flow direction of air

within the blow-by gas recirculation device. The black arrows shown in FIGS. 2 and 3 indicate the flow direction of blow-by gas within the blow-by gas recirculation device.

(23) FIG. 2 is a diagram illustrating the state of the blow-by gas recirculation device during naturally aspirated operation. During naturally aspirated operation, the inside of the intake manifold 28 is at negative pressure, meaning the pressure is lower than atmospheric pressure. The interior of the crankcase 15 is connected to the intake manifold 28 through the first passage R1. The first check valve 42, in the first passage R1, is configured to permit the flow of gas from the interior of the crankcase 15 toward the intake passage 21 through the first passage R1. The interior of the crankcase 15 is connected through the third passage R3 to the portion of the intake passage 21 upstream of the compressor 24. Thus, during naturally aspirated operation, air is introduced into the crankcase 15 through the third passage R3, and the blow-by gas in the crankcase 15 is drawn into the intake manifold 28 through the first passage R1.

(24) FIG. 3 is a diagram illustrating the state of the blow-by gas recirculation device during forced-induction operation. During forced-induction operation, the portion of the intake passage 21 downstream of the compressor 24 is at positive pressure, meaning the pressure is higher than the atmospheric pressure. The second passage R2 connects the interior of the crankcase 15 to the intake manifold 28, which is part of the intake passage 21 at positive pressure. The second check valve 46, in the second passage R2, is configured to permit the flow of gas from the intake passage 21 toward the crankcase 15 through the second passage R2. Thus, during forced-induction operation, air is introduced into the crankcase 15 through the second passage R2. The introduced air at positive pressure causes the blow-by gas to be delivered from the crankcase 15 to the intake passage 21 through the third passage R3.

(25) When a larger amount of gas flows into the crankcase 15 through the second passage R2 than the flow rate of gas delivered to the intake passage 21 through the third passage R3, gas accumulates inside the crankcase 15, causing the internal pressure to increase. When the internal pressure of the crankcase 15 increases, oil blow-up and leakage of blow-by gas to the outside are more likely to occur. In oil blow-up, oil flows into the combustion chamber 17 through the gap between the piston 13 and the cylinder 12. To solve this problem, in the blow-by gas recirculation device of the present embodiment, the minimum flow passage area of the second passage R2 is smaller than the minimum flow passage area of the third passage R3. That is, the third passage R3 is designed to allow a larger amount of gas to flow through than the second passage R2. This limits the increase in the internal pressure of the crankcase 15 that would be caused by the accumulation of gas.

(26) The blow-by gas recirculation device of the present embodiment achieves the following advantages.

(27) (1) The blow-by gas recirculation device of the present embodiment includes the first passage R1, the second passage R2, the third passage R3, the first check valve 42, and the second check valve 46. The first passage R1 connects the portion of the intake passage 21 downstream of the throttle valve 27 to the interior of the crankcase 15. The second passage R2 connects the portion of the intake passage 21 downstream of the intercooler 26 to the interior of the crankcase 15. The third passage R3 connects the portion of the intake passage 21 upstream of the compressor 24 to the interior of the crankcase 15. The first check valve 42 permits the flow of gas from the interior of the crankcase 15 to the intake passage 21 through the first passage R1, and restricts the flow of gas from the intake passage 21 to the interior of the crankcase 15 through the first passage R1. The second check valve 46 permits the flow of gas from the intake passage 21 to the interior of the crankcase 15 through the second passage R2, and restricts the flow of gas from the interior of the crankcase 15 to the intake passage 21 through the second passage R2. The blow-by gas recirculation device ventilates blow-by gas within the crankcase 15 during both naturally aspirated operation and forced-induction operation of the forced-induction engine 10.

(28) (2) If the intake-side end of the second passage R2 were connected to the portion of the intake

passage **21** upstream of the intercooler **26** and downstream of the compressor **24**, it would allow air to be introduced into the crankcase **15** through the second passage **R2** during forced-induction operation. However, in this case, high-temperature air before being cooled by the intercooler **26** would be introduced into the crankcase **15**. By contrast, in the blow-by gas recirculation device of the present embodiment, the second passage **R2** is configured to connect the portion of the intake passage **21** downstream of the intercooler **26** to the interior of the crankcase **15**. Thus, the blow-by gas recirculation device of the present embodiment allows the air cooled by the intercooler **26** to be introduced into the crankcase **15** during forced-induction operation. Accordingly, since dense air is introduced at relatively low temperatures, blow-by gas in the crankcase **15** is ventilated more efficiently compared to when low-density air is introduced at relatively high temperatures.

(29) (3) In the blow-by gas recirculation device of the present embodiment without the second passage **R2** and the second check valve **46**, the third passage **R3** would be used solely as a passage to introduce air into the crankcase **15**. By contrast, the blow-by gas recirculation device of the present embodiment includes the second passage **R2**, which connects the portion of the intake passage **21** downstream of the intercooler **26** to the interior of the crankcase **15**. Further, the blow-by gas recirculation device of the present embodiment includes the second check valve **46**. During forced-induction operation, the second check valve **46** opens and uses the second passage **R2** as an air introduction passage. The second check valve **46** closes during naturally aspirated operation, thereby sealing the second passage **R2**. The arrangement of the second passage **R2** and the second check valve **46** causes the third passage **R3** to function as an air intake passage during naturally aspirated operation and function as a blow-by gas release passage during forced-induction operation. Thus, with a relatively simple configuration, the blow-by gas recirculation device ventilates blow-by gas inside the crankcase **15** during both naturally aspirated operation and forced-induction operation.

(30) (4) In the present embodiment of the blow-by gas recirculation device, during forced-induction operation, air is introduced into the crankcase **15** through the second passage **R2**, and blow-by gas inside the crankcase **15** is released into the intake passage **21** through the third passage **R3**. If the amount of air introduced into the crankcase **15** is greater than the amount of blow-by gas released from the crankcase **15**, the internal pressure of the crankcase **15** increases. In the blow-by gas recirculation device of the present embodiment, the second passage **R2** has a minimum flow passage area smaller than the minimum flow passage area of the third passage **R3**. This makes it easier for gas to flow through the third passage **R3** than through the second passage **R2**. This prevents situations in which gas accumulates in the crankcase **15** and its internal pressure rises during forced-induction operation.

(31) (5) In the blow-by gas recirculation device of the present embodiment, separators used to separate oil mist are disposed in the first passage **R1** and the third passage **R3**, each of which functions as a passage for releasing blow-by gas during natural aspiration operation and forced-induction operation. This prevents oil from mixing with intake air that results from blow-by gas recirculation.

(32) (6) In the present embodiment of the blow-by gas recirculation device, the second check valve **46** is located at the portion of the second passage **R2** connected to the crankcase **15**. The second check valve **46** is configured to restrict the flow of gas from the interior of the crankcase **15** to the second PCV hose **45**. Thus, even if the second PCV hose **45** is torn or disconnected from a connector to damage the second passage **R2**, the blow-by gas inside the crankcase **15** is less likely to be released into the external air. Further, if the second passage **R2** is damaged, the intake manifold **28** is exposed to the external air. When the intake manifold **28** is exposed to the external air, the intake pressure and intake flow rate change immediately. Accordingly, the arrangement of the second check valve **46** at the portion of the second passage **R2** connected to the crankcase **15** allows for easy and quick detection of damage to the second passage **R2**.

(33) (7) The function of the first check valve **42** is similar to that of a PCV valve used in the blow-

by gas recirculation device for a naturally aspirated engine. This allows an existing PCV valve to be employed in the first check valve **42**. Further, the functions of the first check valve **42** and the second check valve **46** are identical except for the direction in which they permit or restrict the flow of gas. This allows the first check valve **42** and the second check valve **46** to be used as common components. Alternatively, many aspects of the design of the first check valve **42** are applicable to the design of the second check valve **46**.

(34) Modifications

(35) The present embodiment may be modified as follows. The present embodiment and the following modifications can be combined as long as they remain technically consistent with each other.

(36) In the above embodiment, the intake-side end of the second passage **R2** is connected to the intake manifold **28**. The intake-side end of the second passage **R2** may be connected to a portion other than the intake manifold **28** if it is a portion of the intake passage **21** downstream of the intercooler **26**. For example, the intake-side end of the second passage **R2** may be connected to the section of the intake passage **21** between the intercooler **26** and the throttle valve **27**.

(37) The third passage **R3** may be configured to connect the interior of the crankcase **15** to the intake passage **21** without passing through the valve operating chamber **20**. For example, FIG. **4** shows an example of a third passage **R3** formed by a third PCV hose **52** that connects the crankcase **15** to the intake passage **21**. One end of the third PCV hose **52** is connected to the cylinder block **11**, and the other end is connected to a portion of the intake passage **21** located between the air cleaner **23** and the compressor **24**. In the forced-induction engine **10** shown in FIG. **4**, a second separator **53** that separates oil mist is disposed in the middle of the third PCV hose **52**. The second separator **53** is fixed to the outer wall of the cylinder block **11**.

(38) The second check valve **46** may be located in the portion of the second passage **R2** connected to the intake passage **21** or in the middle of the second passage **R2**.

(39) The second passage **R2** may be configured to connect the interior of the crankcase **15** to the intake passage **21** through the valve operating chamber **20**. FIG. **4** illustrates an example configuration of such a second passage **R2**. The second passage **R2** in FIG. **4** connects the interior of the crankcase **15** to the intake manifold **28** through the oil return passage **47**, the valve operating chamber **20**, the second PCV hose **50**, and the second check valve **46**. The opposite ends of the second PCV hose **50** in FIG. **4** are respectively connected to the head cover **16A** and the intake manifold **28**. In FIG. **4**, the second check valve **46** is located at a portion of the second PCV hose **50** connected to the intake manifold **28**.

(40) In the blow-by gas recirculation device in the above embodiment, the two separators (i.e., head-side separator **41** and block-side separator **44**) are disposed as the first separator to separate oil mist from the gas flowing through the first passage **R1**. The number of separators included in the first separator may be changed. Additionally, the second separators **48**, **53**, which separate oil mist from the gas flowing through the third passage **R3**, may include multiple separators. If burning the oil mist in the blow-by gas in the combustion chamber **17** is sufficient for treatment, those separators may be omitted.

(41) The second passage **R2** may have a minimum flow passage area equal to or larger than that of the third passage **R3**. In such a case, the increase in oil blow-up, the leakage of blow-by gas, and the like resulting from internal pressure rise in the crankcase **15** are limited by disposing an additional mechanism in the forced-induction engine **10** to limit the internal pressure rise.

(42) Some or all of the first PCV hose **43**, the second PCV hoses **45**, **50**, and the third PCV hoses **49**, **52** may be replaced with metal hoses.

(43) The blow-by gas recirculation device of the above embodiment may be employed in forced-induction engines other than hydrogen engines, such as gasoline engines and diesel engines.

(44) For an engine including banks and compressors, such as a V-type engine, the blow-by gas recirculation device of the above embodiment may be disposed in each bank.

(45) Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A blow-by gas recirculation device configured to recirculate blow-by gas from an interior of a crankcase of a forced-induction engine to an intake passage, wherein the forced-induction engine includes: a compressor disposed in the intake passage; an intercooler disposed in a portion of the intake passage downstream of the compressor; and a throttle valve disposed in a portion of the intake passage downstream of the intercooler, the blow-by gas recirculation device comprises: a first passage that connects a portion of the intake passage downstream of the throttle valve to the interior of the crankcase; a second passage that connects the portion of the intake passage downstream of the intercooler to the interior of the crankcase; a third passage that connects a portion of the intake passage upstream of the compressor to the interior of the crankcase; a first check valve configured to permit a flow of gas from the interior of the crankcase to the intake passage through the first passage and restrict a flow of gas from the intake passage to the interior of the crankcase through the first passage; and a second check valve configured to permit a flow of gas from the intake passage to the interior of the crankcase through the second passage and restrict a flow of gas from the interior of the crankcase to the intake passage through the second passage, and the second passage and the third passage are separate from each other.
 2. The blow-by gas recirculation device according to claim 1, wherein a minimum flow passage area of the second passage is smaller than a minimum flow passage area of the third passage.
 3. The blow-by gas recirculation device according to claim 1, comprising: a first separator configured to separate oil mist from the gas flowing through the first passage; and a second separator configured to separate oil mist from the gas flowing through the third passage.
 4. The blow-by gas recirculation device according to claim 3, wherein the forced-induction engine includes a cylinder block including the crankcase, a cylinder head mounted on an upper portion of the cylinder block, and a head cover mounted on an upper side of the cylinder head, and the first passage connects the intake passage to the interior of the crankcase via the head cover, the cylinder head, and the cylinder block.
 5. The blow-by gas recirculation device according to claim 4, wherein the first separator includes a head-side separator provided in the head cover and a block-side separator provided in the cylinder block.
 6. The blow-by gas recirculation device according to claim 1, wherein the second check valve is located at a portion of the second passage connected to the crankcase.
 7. The blow-by gas recirculation device according to claim 1, wherein a fuel for the forced-induction engine is hydrogen.
 8. The blow-by gas recirculation device according to claim 1, wherein the second passage connects the portion of the intake passage downstream of the throttle valve to the interior of the crankcase.
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